

Program : Diploma in Fire Technology and Safety	
Course Code: 3461	Course Title: Industrial Safety Engineering
Semester: 3	Credits: 3
Course Category: Program Core	
Periods per week: 3 (L:2 T:1 P:0)	Periods per semester: 45

Course Objectives:

- To understand the fundamental principles of industrial safety and accident prevention.
- To recognize the significance of safety management and its role in reducing accidents.
- To learn various safety planning and organizational behavior techniques.
- To analyze safety education, training methods, and employee participation in promoting safety.

Course Prerequisites:

Topic	Course code	Course name	Semester
Principles of Safety Management		Principles of Safety Management	2

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Identify the principles of accident prevention management and its application in industrial safety.	12	Understanding
CO2	Analyze planning and safety strategies and their impact on organizational behavior.	11	Applying
CO3	Organize safety programs and direct the implementation of safety measures in industries.	11	Understanding
CO4	Apply safety education, training, and employee participation in safety promotion schemes.	9	Understanding

	Series Test	2	
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CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2	3	2			
CO2	3	3	2	2			
CO3	3	3	2	3			
CO4	3	3	3	2			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Introduction to the significance and evolution of industrial safety and accident prevention.		
M1.01	Introduce industrial safety principles and their historical evolution.	6	Understanding
M1.02	Explain safety management principles and practices.	6	Understanding
CO2	Understand safety planning techniques and how human behavior influences safety management.		
M2.01	Analyze safety planning methods and objectives.	5	Understanding
M2.02	Explain accident data analysis and remedial measures.	6	Understanding
	Series Test – I	1	
Contents: Planning for Safety & Organizational Behavior and Safety Safety planning methods and objectives. Accident data analysis techniques and their role in creating remedial measures. Human behavior and its influence on safety. Motivation theories related to safety practices.			
CO3	Learn about organizing and directing safety programs within industrial environments.		

M3.01	Identify the organizational structure needed for implementing safety measures.	5	Understanding
M3.02	Understand the formation and functions of safety committees and departments.	6	Understanding
Contents: Organizing for Safety & Directing for Safety Organizational structure for safety in industries. Formation and roles of safety committees. Functions and importance of safety departments within organizations			
CO4	Understand and apply safety education, training methods, and employee participation strategies in promoting safety.		
M4.01	Discuss safety education and the importance of safety training methods.	6	Understanding
M4.02	Explain methods of employee participation in safety programs.	3	Understanding
	Series Test – II	1	
Contents: Safety Education and Training & Employee Participation in Safety Training methods for safety and assessing training needs. Designing effective safety education programs. Employee participation in safety measures and safety promotion. Safety incentive schemes and employee motivation in safety practices.			

Text / Reference:

T/R	Book Title/Author
T1	Industrial Safety Management - R K Jain
T2	Industrial Safety - P. K. Gupta
R1	Introduction to Industrial Safety - H. K. Gupta
R2	Industrial Safety Engineering - K. R. Prabhakaran
R3	Safety Management in Industry - James R. Wetherbee

Online Resources:

Sl.No	Website Link
1	https://www.osha.gov/
2	https://www.niosh.gov/
3	https://www.safetyxchange.org/
4	https://nptel.ac.in/courses/116/105/116105081/